INDIVIDUAL PROGRESS REPORT

Task 9: Integrated Pipeline, Algorithm Metrics, and Professional Visualizations

Group Topic	Boardie Analytics: A Big Data Approach to Student Boarding House Discovery and Geofenced Presence Pattern Analysis
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Lab No.	Task 9
Git Repo/Collab Link	https://github.com/carlpoot/week-4-biggs-data

Objectives

The primary goal of Task 9 is to expand upon the previous Boardie Analytics algorithm, transforming it into a comprehensive analytics pipeline capable of integrating with third-party applications and generating professional-grade visual outputs. This phase centers on structuring data for dashboards, extracting model performance metrics, building visual representations, and organizing all deliverables for GitHub deployment.

Tasks Completed

- Evaluated the Task 9 instructions and linked the new requirements to the Random Forest classification model developed in Week 3.
- Developed a cohesive analytics pipeline that imports the cleaned Boardie dataset and generates new features for dashboards, such as review quality, availability status, listing size, and amenity tiers.
- Advanced the classification model by training a Random Forest algorithm to predict price categories using inputs like room details, availability, reviews, and property traits.
- Optimized model performance using GridSearchCV, adjusting key parameters like maximum depth, minimum samples per leaf, and the number of estimators (trees).
- Extracted key algorithm evaluation metrics, including accuracy, weighted F1 score, weighted precision, weighted recall, a confusion matrix, and a classification report.
- Generated processed data files formatted for external BI tools, specifically a dashboard-optimized CSV ready for Tableau, Looker Studio, Power BI, or spreadsheet integration.
- Designed professional visual charts, including feature importance, price band distribution, availability and room types by price band, a confusion matrix, association-rule lift charts, and median prices by neighborhood.
- Incorporated a basic association-pattern analysis utilizing support, lift, and confidence metrics to reflect Chapter 9's focus on association rule mining.
- Constructed an HTML-based browser dashboard prototype to illustrate how the metrics

and visuals can be professionally displayed.
- Compiled the Python script, notebook, processed CSVs, visual assets, README, and this progress report for submission to GitHub.

Tasks to be Completed

- Push the Task 9 script, notebook, generated visuals, CSV outputs, and progress report to the team's GitHub repository.
- Update the repository link in this document once the files are successfully uploaded.
- If mandated by the instructor, load the formatted CSV into Tableau (or another BI platform) to construct a fully interactive dashboard.
- Further enhance the machine learning model by experimenting with new algorithms or integrating additional contextual features if the dataset grows.

Pipeline Summary

Pipeline Stage	**Output / Purpose**
Feature Preparation	Ingests the processed Week 3 Boardie dataset to engineer dashboard and model-ready fields (e.g., amenity level, review quality, availability label).
Model Training	Executes a Random Forest Classifier to predict property price tiers.
Algorithm Metrics	Computes and exports accuracy, weighted precision, weighted recall, weighted F1 score, confusion matrix, and classification report.
External Tool Integration & Professional Visualizations	Outputs tableau_boardie_dashboard_data.csv and other summaries for BI tools. Generates presentation-ready PNG visuals and an HTML dashboard_preview.html file.

Algorithm Metrics Summary

Metric	Value
Accuracy	0.7200
Weighted Precision	0.7278
Weighted Recall	0.7200
Weighted F1 Score	0.7049
Training Rows	750
Testing Rows	250
Best Parameters	{"model_max_depth": 10, "model_min_samples_leaf": 1, "model_n_estimators": 100}

Results and Analysis

Task 9 successfully produced actionable model evaluations and professional-grade visualizations. To categorize property listings into distinct price bands, a Random Forest Classifier was utilized, leveraging availability and listing attributes as predictive inputs.

Following hyperparameter optimization, the model reached a 0.7200 accuracy rate and a 0.7049 weighted F1 score. This establishes a solid baseline for predicting price categories, despite the relative scarcity of budget and premium listings compared to the dominant mid-range and upper-mid tiers.

The exported CSV files highlight the integration focus of this task. Instead of keeping results confined to a notebook, these datasets are structured for seamless ingestion into external platforms like Power BI, Tableau, Looker Studio, or Excel.

These visualization diagrams represent the analytical findings. The room type and price range charts illustrate listing distributions, while the feature importance and availability graphs reveal the underlying factors driving the model's predictions. The confusion matrix highlights where the model succeeded and where it misclassified properties. Additionally, the association pattern findings align with Chapter 9 concepts by showcasing co-occurrence rules via confidence, support, and lift metrics.

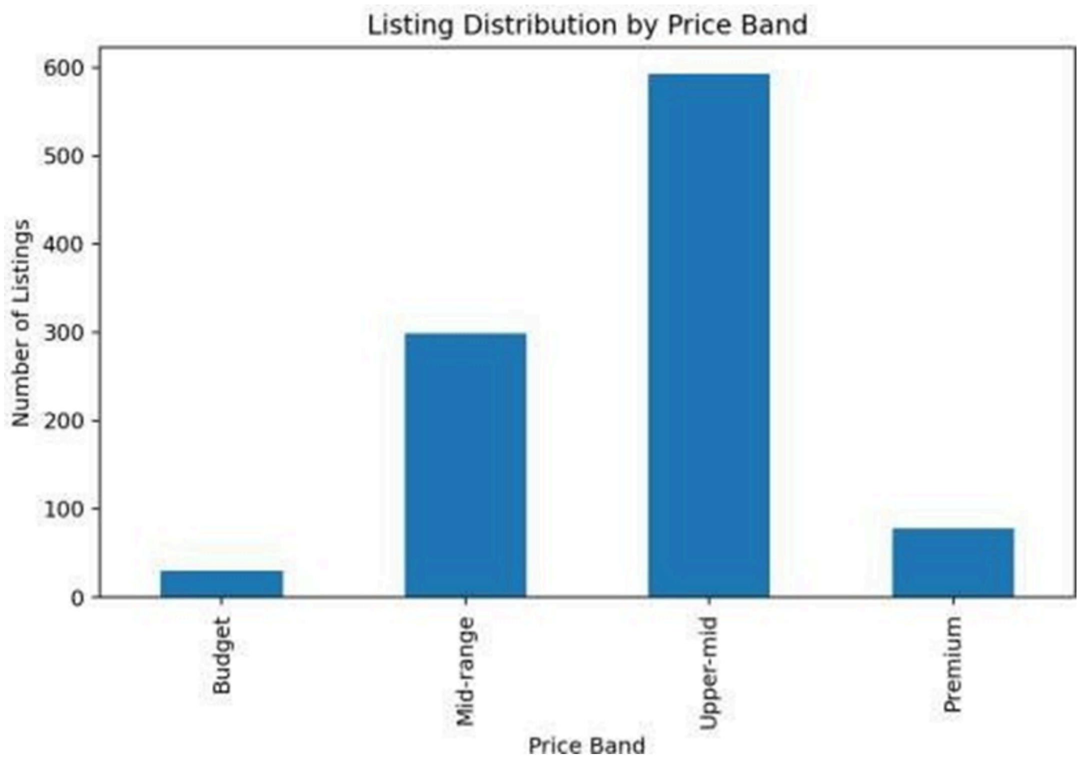

Figure 1. Listing distribution by price band

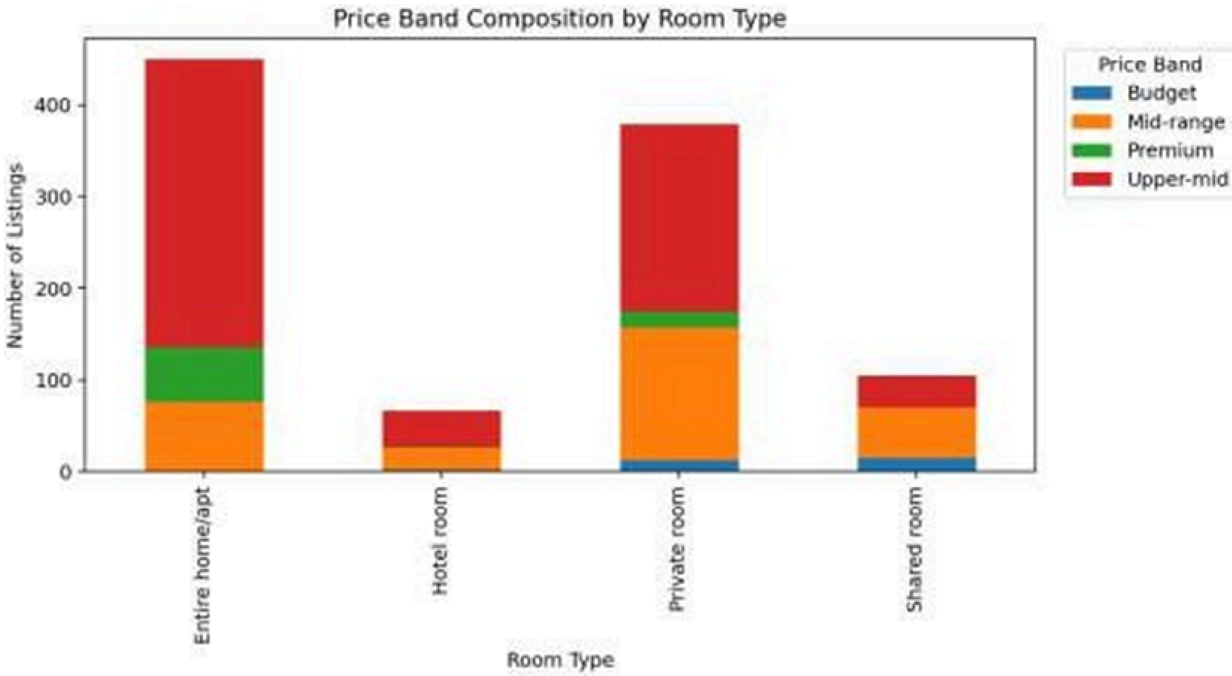

Figure 2. Price band composition by room type

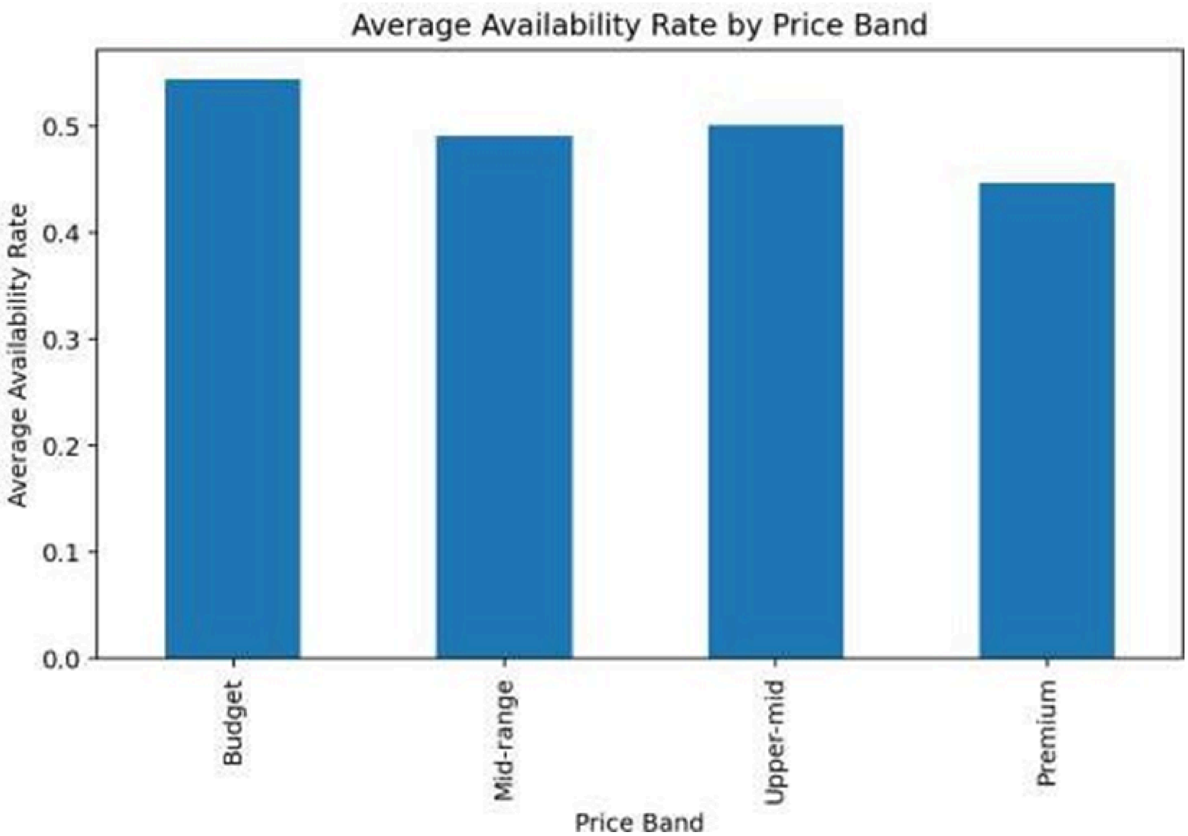

Figure 3. Average availability rate by price band

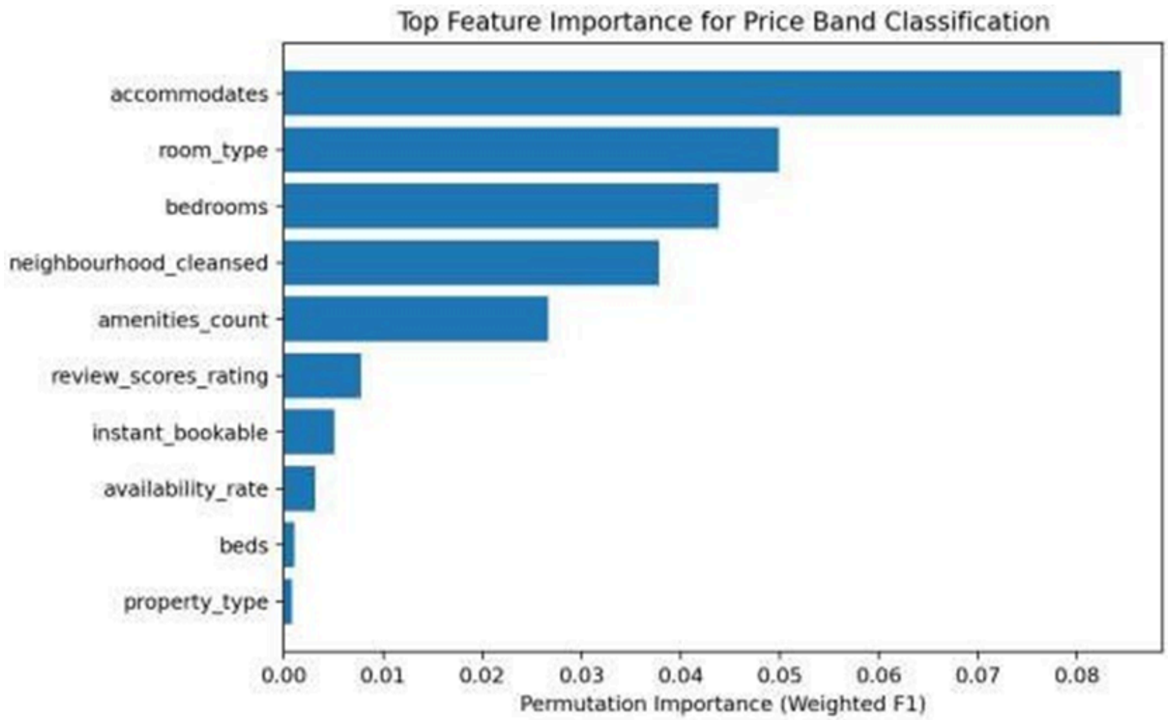

Figure 4. Feature importance for price-band classification

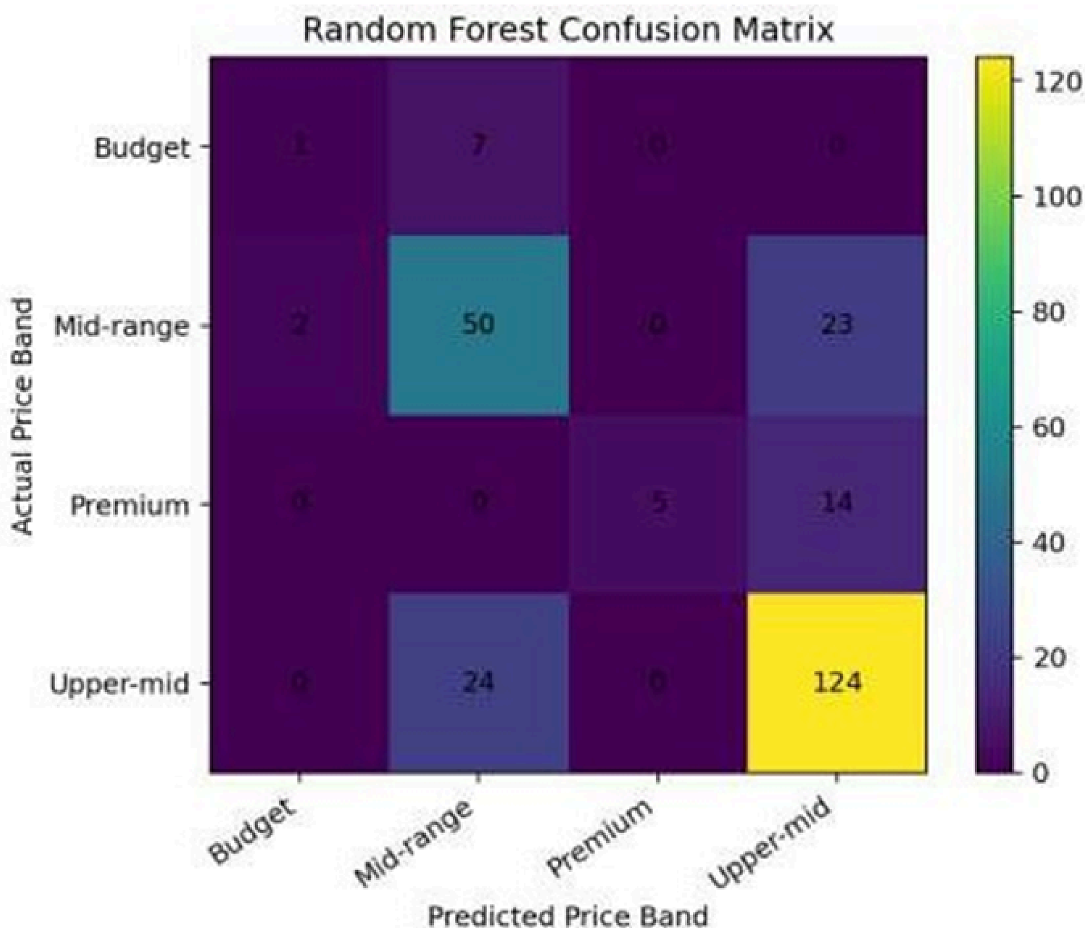

Figure 5. Random Forest confusion matrix

Figure 6. Top areas by median listing price

Challenges and Solutions

Challenge	**Solution**
The assignment mandates external tool integration, but the team might not have a finished Tableau workbook yet.	The pipeline generates an HTML dashboard prototype and dashboard-optimized CSVs for future BI tool ingestion.
Uneven distribution of price-band classes can skew classification accuracy.	Implemented class-weight balancing and stratified train-test splitting to mitigate the impact of minority classes.
Professional visualization demands more than standard exploratory graphs.	Produced refined charts featuring proper labels, titles, and summaries, alongside a presentation-ready browser preview.

Previous tasks focused on classification, but Chapter 9 covers unsupervised association rule mining.	Integrated a foundational association-pattern module calculating lift, support, and confidence, while retaining the Random Forest classification metrics.

Conclusion

Task 9 successfully transformed the existing Boardie Analytics Model into a comprehensive analytics pipeline. Deliverables included refined classification outputs, performance metrics, BI-ready data exports, professional charts, an HTML dashboard mockup, and this status report. Ultimately, this exercise highlights the process of migrating processed machine learning data from an internal coding environment to external, professional visualization platforms.